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STRONG ASSOCIATIONS OF SHORT-CHAIN PERFLUOROALKYL ACIDS WITH SERUM ALBUMIN AND INVESTIGATION OF BINDING MECHANISMS

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Abstract—Interactions of perfluoroalkyl acids (PFAAs) with tissue and serum proteins likely contribute to their tissue distribution and bioaccumulation patterns. Protein–water distribution coefficients (K_{PW}) based on ligand associations with bovine serum albumin (BSA) as a model protein were recently proposed as biologically relevant parameters to describe the environmental behavior of PFAAs, yet empirical data on such protein binding behavior are limited. In the present study, associations of perfluoroalkyl carboxylates (PFCAs) with two to 12 carbons (C_2–C_{12}) and perfluoroalkyl sulfonates with four to eight carbons (C_4, C_6, and C_8) with BSA are evaluated at low PFAA:albumin mole ratios and various solution conditions using equilibrium dialysis, nanoelectrospray ionization mass spectrometry, and fluorescence spectroscopy. Log K_{PW} values for C_4 to C_{12} PFAAs range from 3.3 to 4.3. Affinity for BSA increases with PFAA hydrophobicity but decreases from the C_8 to C_{12} PFCAs, likely due to steric hindrances associated with longer and more rigid perfluoroalkyl chains. The C_4-sulfonate exhibits increased affinity relative to the equivalent chain-length PFCA. Fluorescence titrations support evidence that an observed dependence of PFAA-BSA binding on pH is attributable to conformational changes in the protein. Association constants determined for perfluorobutanesulfonate and perfluoropentanoate with BSA are on the order of those for long-chain PFAAs ($K_a \sim 10^6$/M), suggesting that physiological implications of strong binding to albumin may be important for short-chain PFAAs. Environ. Toxicol. Chem. 2011;30:2423–2430. © 2011 SETAC

Keywords—Protein binding Bioaccumulation Modeling Perfluoroalkyl acids Albumin

INTRODUCTION

The unique chemical properties of perfluoroalkyl acids (PFAAs), a class of stable anionic surfactants, have been capitalized on since the 1940 s in the production of a variety of industrial and consumer products [1]. In response to comprehensive research documenting the environmental persistence and widespread occurrence of PFAAs in humans and wildlife [2,3], the 3M Company voluntarily eliminated perfluorooctane sulfonyl fluoride–based materials including perfluorooctanoate (PFOA) and perfluorooctane sulfonate (PFOS) from production. Eight companies subsequently committed to eliminating emissions of PFOA and related compounds by 2015 as part of a U.S. Environmental Production Agency PFOA Stewardship Program [4]. Internationally, however, perfluoroalkyl compounds of varying chain lengths (C_4–C_{15}) are still manufactured, and legacy products remain in use globally [1]. The phase-out of PFOS has been accompanied by a shift in production to shorter-chain-length compounds, including those based on C_4-sulfonyl chemistries [4]. The apparently efficient clearance of perfluorobutane sulfonate (PFBS), observed in several organisms, reduces this compound's bioaccumulative potential [5,6].

The structures of two homolog groups, perfluoroalkyl carboxylates (PFCAs) and perfluoroalkyl sulfonates (PFSAs), resemble those of fatty acids and hydrocarbon-based detergents (Supplemental Figure S1), but the high-energy carbon–fluorine bond imparts resistance to hydrolysis, photolysis, microbial degradation, and metabolism by vertebrates and renders the perfluoroalkyl tail both hydrophobic and oleophobic [7,8]. Rather than partitioning to adipose tissue, PFAAs are detected predominantly in protein-rich compartments such as the liver, kidney, and blood [9–12], and PFOS concentrations have been shown to positively correlate with tissue and fluid protein content [9].

Recent field monitoring data for C_7 to C_{14} PFCAs demonstrate a curvilinear relationship between protein-normalized trophic magnification factors in a marine mammalian food web and protein–water distribution coefficients (K_{PW}) estimated from a relationship between octanol–water partition coefficients and affinities for bovine serum albumin (BSA) [13]. The K_{PW} may be incorporated into predictive models to improve estimations of chemical distribution and bioaccumulation of organic contaminants [14]. A globular protein consisting of 583 amino acid residues, BSA (66,430 Da) is widely used as a model protein in biophysical, biochemical, and physicochemical studies, because BSA binding sites accommodate a wide variety of endogenous and exogenous ligands. Albumin is located to some degree in every fluid of the body, accounting for 60% of total serum proteins at concentrations of 35 to 50 g/L and exhibiting extravascular concentrations of 10 to 30 g/L in the skin, muscle, liver, gut, and subcutaneous compartment [15]. Despite a proposal that PFAA biomagnification patterns may correlate with binding to serum proteins, empirical data on the protein-binding behavior of PFAAs are limited, especially for short-chain PFAAs. Chen and Guo [16] measured displacement of fluorescent probes on human serum albumin (HSA) to calculate association constants for PFBS and the C_4 PFCA on the order of 10^6/M. However, other commonly utilized fluorescence methods may not be applicable to the study of short-chain PFAAs with albumin [17,18].

All Supplemental Data may be found in the online version of this article.

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A wide range of association constants (10^2–10^6/M) for C_8 and C_9 PFAAs with rat [12], human [12,16,17,19–22], and bovine [18,22,23] albumins are reported. Primary association constants for PFOA and perfluorononanoate (PFNA) with BSA suggest binding through specific high-affinity interactions. High log K_{PW} values (>4) were determined for PFOA and PFNA at physiological PFAA:albumin mole ratios [23]. Site-specific binding of PFOA and perfluorohexanoate (PFHxA) to albumin has the potential to disrupt endogenous functions via displacement of the fatty acid oleate [24], which exhibits 10^7 to 10^8/M association constants with albumin for its first five binding sites [25]. However, C_4 to C_{10} PFAAs did not displace steroid hormones from avian serum proteins at physiological concentrations [26]. In addition to active uptake of PFAAs by transporter systems [27], PFAA–albumin binding may play a role in elimination kinetics [24]. Binding of small molecules to albumin controls their free concentrations in blood and influences the duration of action, because generally only the unbound fraction can exit the vascular component. Interactions of PFAAs with other biomolecular targets including liver fatty acid–binding proteins also may play a role in PFAA bioaccumulation [28] and affect the pharmacokinetics of fatty acids or other endogenous ligands [29].

The objective of the present study was to examine the interactions of a series PFCAs (C_5–C_{12}) and even-chain-length PFSAs (C_4–C_8) with BSA. The fraction of PFAAs bound to BSA, and protein–water distribution coefficients are determined at physiologically relevant ligand:protein mole ratios, using equilibrium dialysis and liquid chromatography tandem mass spectrometry, a direct approach that allows evaluation of binding parameters at low ligand concentrations. Further analyses through dialysis, nanoelectrospray ionization mass spectrometry (nanoESI-MS), and fluorescence spectroscopy offer insights into the mechanisms of PFAA–albumin interactions by evaluating various solution- and chemical structure–specific parameters potentially affecting the binding of PFAAs to albumin. In particular, the influence of fluorocarbon carbon chain length, ionic head group, and solution pH and ionic strength are assessed.

METHODS

Equilibrium dialysis

A purity-corrected equimolar stock solution of PFCAs (C_5–C_{12} and C_{14}) and PFSAs (C_4, C_6, and C_8) and a separate concentrated solution of PFNA were prepared in high-performance liquid chromatography (HPLC) grade water and diluted in 50 mM sodium phosphate buffer in polypropylene containers without the use of an organic co-solvent. BSA solutions were prepared fresh daily in glass volumetric flasks, using 50 mM sodium phosphate buffer at the desired pH, and concentrations were measured using absorbance readings from a NanoDrop-1000 spectrophotometer (NanoDrop Technologies). Polypropylene dialysis reservoirs containing the same buffers (250–500 ml) and spiked with PFAAs were sampled in triplicate to determine initial (0 h) and final (120 h) PFAA concentrations. The buffer system was selected for consistency with previous analyses [22,23]. Dialysis reservoir pH and temperature measurements were taken with an Orion 5 Star Multimeter (Thermo Fisher Scientific). Dialysis bags containing 2 ml BSA solution were secured with dialysis clips, added to reservoirs, and equilibrated for 120 h at room temperature ($21.2 \pm 0.4^{\circ}C$) before sampling. Six reservoirs were prepared at pH 7.0 with a range of mixed PFAA concentrations and one dialysis bag each ([BSA] = 199 ± 4 μM) and were sampled in triplicate at equilibrium to determine K_{PW} values. In these tests, individual free PFAA concentrations at equilibrium ranged from 0.009 to 0.7 μM (Table 1), such that total free PFAA concentrations ranged from 0.8 to 5 μM. Additional dialysis reservoirs prepared at pH 6.1, 7.0, 8.0, and 8.9 with 2.8 ± 0.6 μM total PFAAs each contained three dialysis bags ([BSA] = 197 ± 2 μM) that were sampled in duplicate for PFAAs. Over the pH range studied, the PFAAs tested were in ionized form in solution [30]; any changes in protonation state on binding to BSA are not detectable by this method. Unless otherwise stated, error bars represent 1 standard deviation. Control dialysis bags containing a buffer-only solution were equilibrated in a PFAA-spiked reservoir to confirm equilibration and diffusion of PFAAs through the dialysis membrane. Control results indicate that perfluorotetradecanoic acid does not diffuse freely into the dialysis bag, so results for perfluorotetradecanoic acid were excluded.

Details of postdialysis sample preparation, PFAA analysis via liquid chromatography tandem mass spectrometry, and mass balance results are available in the Supplemental Data (Table S1). Samples and standards were analyzed using a Shimadzu LC system (LC10ADvp pumps controlled by an SCL10Avp controller) coupled to a Sciex API 3000 triple quadrupole mass spectrometer (MDS Sciex) operating in negative electrospray ionization multiple reaction monitoring mode. Analyte separation was achieved using a 40 mm × 2.1 mm Targa Sprite C18 column (5-μm particle size, Higgins Analytical) with a C18 guard column (Higgins Analytical).

Table 1. Fraction of perfluoroalkyl acids (PFAAs) bound to 200 μM bovine serum albumin (BSA) and log protein–water distribution coefficients for PFAAs with BSA measured over a range of equilibrium free PFAA reservoir concentrations[a]

Analyte	Reservoir [PFAA] Range (μM)[b]	Log K_{PW}	Fraction bound
Perfluorobutanesulfonate (PFBS)	0.010 to 0.57	3.86 ± 0.07	$99.0 \pm 0.5\%$
Perfluorohexanesulfonate (PFHxS)	0.012 to 0.27	4.3 ± 0.1	$99.2 \pm 0.5\%$
Perfluorooctanesulfonate (PFOS)	0.013 to 0.26	4.1 ± 0.1	$99.1 \pm 0.4\%$
Perfluoropentanoate (PFPeA)	0.024 to 0.56	3.40 ± 0.02	$96.6 \pm 0.8\%$
Perfluorohexanoate (PFHxA)	0.009 to 0.43	4.05 ± 0.02	$99.2 \pm 0.3\%$
Perfluoroheptanoate (PFHpA)	0.054 to 0.30	4.23 ± 0.08	$99.3 \pm 0.1\%$
Perfluorooctanoate (PFOA)	0.023 to 0.32	4.14 ± 0.04	$99.1 \pm 0.2\%$
Perfluorononanoate (PFNA)	0.064 to 0.46	4.05 ± 0.08	$98.9 \pm 0.3\%$
Perfluorodecanoate (PFDA)	0.016 to 0.47	3.86 ± 0.08	$98 \pm 1\%$
Perfluoroundecanoate (PFUnA)	0.026 to 0.38	3.7 ± 0.2	$95 \pm 3\%$
Perfluorododecanoate (PFDoA)	0.031 to 0.68	3.3 ± 0.1	$80 \pm 10\%$

[a] For the fraction of PFAA bound to BSA in each tested reservoir, errors represent 95% confidence values; standard errors for log K_{PW} are from regressions performed using Kaleidagraph software.
[b] $n = 6$; $n = 5$ for PFHpA.

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Nanoelectrospray ionization mass spectrometry

Sample preparation, as well as instrument parameters and conditions, were previously described [23]. Briefly, individual PFAAs prepared in 9 mM ammonium acetate (pH 7) at room temperature were added to BSA (50 μM) in polypropylene microcentrifuge tubes, brought to a 1:1 or 2:1 PFAA:BSA mole ratio, and equilibrated for one or more hours before same-day analysis. Samples were analyzed using an Advion Triversa Nanomate nano-electrospray robot (Advion BioSystems) coupled with a Waters Micromass Q-Tof API-US quadrupole time-of-flight mass spectrometer. Instrument gas pressure and voltage settings were selected to maximize peak intensities while maintaining resolution and ion current. For static collision energy tests, mass spectra were acquired for 4 min with cone voltage set at 100 V and laboratory scale collision energies set to 10 eV for singly charged ions. In duplicate collision-induced dissociation tests for each tested PFAA–BSA complex, the collision energy was ramped from 10 to 90 eV, with spectra acquired for 2 min at each setting. Additional details are available in the Supplemental Data.

Fluorescence spectroscopy

The PFAA stock solutions were prepared in 50-ml polypropylene centrifuge tubes (Corning) in HPLC grade water and were sonicated at 35°C to dissolve. Solutions of BSA were prepared in glass volumetric flasks in 50 mM sodium phosphate buffer (pH 6.1, 7.0, 8.1, or 9.1) immediately before each experiment. A stock solution of sodium chloride (3.31 M) was prepared in HPLC grade water as needed. Fluorescence titrations were conducted for PFNA or PFOS with BSA in solutions buffered at pH 6.1, 7.0, 8.1, or 9.1 without added sodium chloride and at pH 7.0 with added sodium chloride to achieve an ionic strength of 0.21, 0.30, or 0.41 M. Test solutions (14) were prepared in 15-ml polypropylene centrifuge tubes and allowed to equilibrate overnight at 4°C. The concentration of BSA was constant at 4 μM, a concentration that was selected, along with instrumental parameters, to maximize both sensitivity and resolution. One sample was prepared with no added PFAA, whereas the other 13 contained a range of PFAA:BSA mole ratios (0.1:1 through 60:1). Emission scans were collected in a 1-cm quartz cuvette (Starna Cells) using a PTI QuantaMaster spectrofluorometer (Photon Technology International). The excitation wavelength was 295 nm, and the emission was recorded from 305 to 450 nm with a step size of 1 nm and an integration time of 0.1 s. All monochromator slit widths were 4 nm.

RESULTS AND DISCUSSION

Effect of chain length on binding of PFAAs to BSA

Dialysis experiments allow direct measurement of PFAA-BSA binding through partitioning of PFAAs between an external reservoir and BSA-containing dialysis bags. Measured concentrations of C_2 to C_{12} PFCAs and C_4 to C_8 PFSAs in the dialysis bag relative to the external reservoir indicate free movement of PFAAs through the dialysis membrane and retention of PFAA–BSA bound complexes in the dialysis bag. At a BSA concentration of 200 μM (13 g/L) and a range of PFAA mixture concentrations, PFCAs and PFSAs with four to 10 fluorinated carbons were highly bound (>95%) to BSA across the range of free PFAA concentrations tested (Table 1). This was consistent even for the short-chain compounds PFBS, perfluoropentanoate (PFPeA), and PFHxA, for which reduced affinity for BSA was expected. The lower fraction bound for perfluorododecanoate (PFDoA) relative to other compounds in the mixed reservoir could reflect a decrease in affinity at this longer chain length; however, the relatively large error associated with these results and the mass balance calculations limits full interpretation. In a separate reservoir spiked only with PFNA, 98.2% of PFNA was bound to BSA at equilibrium, a result in close agreement with that obtained in the mixed PFAA experiments.

Protein–water distribution coefficients (K_{PW}) were determined for each PFAA by a linear regression of the ratio of analyte concentration in the bound phase (C_p) to that in the aqueous phase (C_w) (Fig. 1). This ratio may be represented by

$$K_{PW} = \frac{C_P}{C_W} = \frac{f_{bound}}{\rho_{albumin} \cdot [P](1-f_{bound})}$$

where [P] is the protein concentration (g/ml), f_{bound} is the fraction bound to protein, and $\rho_{albumin}$ is the partial specific volume of protein in aqueous solution (0.733 ml/g, [31]). The K_{PW} thus has units of (g bound PFAA/ml BSA)/(g free PFAA/ml water) and represents the distribution of the combined anionic and neutral forms of PFAAs between BSA and the aqueous buffer. At the low PFAA concentrations tested, the binding isotherm for total PFAAs, represented by the sum of the concentrations of tested PFAAs, is linear (Fig. S2). Values of log K_{PW} for individual PFAAs range from 3.3 to 4.3 (Table 1). The highest C_w value for PFBS, PFPeA, and PFHxA ($C_w > 0.4$ μM) was excluded in the K_{PW} regressions (Supplemental Data, Table S2) because these binding isotherms were nonlinear at higher concentrations. All data were included in nonlinear regressions performed to determine association constants for the short-chain PFAAs. Data obtained at relatively higher PFAA aqueous concentrations here and in previous work [23] demonstrate an expected nonlinear binding relationship, and illustrate that application of K_{PW} should be limited to a narrow concentration range in which binding may be approximated as linear. At the PFAA:BSA ratios tested, log K_{PW} results were consistently greater than or at the high end of the range of BSA–water distribution coefficients compiled for traditional hydrophobic organic contaminants (log $K_{PW} = 0.09$–3.5) [14].

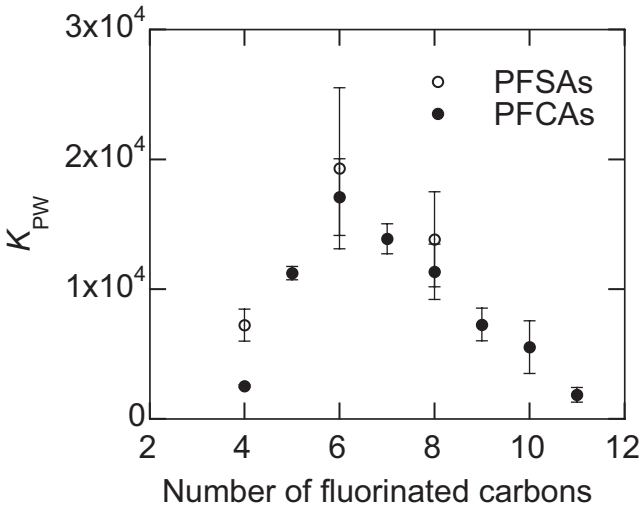


Fig. 1. Measured BSA–water distribution coefficients (K_{PW}) for perfluoroalkyl sulfonates (PFSAs, white circles) and perfluoroalkyl carboxylates (PFCAs, black circles) with fluorocarbon tail lengths of 4 to 11. Error bars are standard errors for regressions performed in Kaleidagraph software.

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An increase in K_{PW} with increasing chain length was observed for PFCAs with 4 to 6 fluorinated carbons. For PFCAs with more than six fluorinated carbons, K_{PW} values generally decreased, although the differences between 6 and 7, 7 and 8, and 9 and 10 fluorinated carbons were not significant. Kelly et al. [13] estimated K_{PW} values for PFCAs with BSA based on a generalized BSA–ligand relationship with octanol–water partition coefficients (K_{OW}). Reported log K_{PW} values increased linearly from 2.0 for PFHpA to 5.0 for PFDoA; however, details of the K_{PW} calculation were not provided. The trend predicts an increase in affinity for BSA with increased hydrophobicity of the perfluoroalkyl tail but does not account for increased steric hindrance associated with longer fluorocarbon tails. Increased rigidity of the fluorocarbon tail and changes in molecular geometry that begin at C_8 and become more pronounced for PFCAs with 10 carbons and longer may influence PFAA partitioning and binding behavior [24,32]. Hydrophobic binding cavities in BSA may have limited ability to accommodate these larger, less flexible ligands.

Direct binding observed by nanoESI-MS

NanoESI-MS, employing a soft ionization technique, is used here to confirm direct binding of PFCAs (C_2–C_9) and PFSAs (C_4–C_8) to BSA and determine stoichiometries of binding at 1:1 PFAA:albumin mole ratios. Nanoelectrospray ionization mass spectrometry has been widely used to study proteins in their native conformation and noncovalent interactions of protein–ligand complexes preserved in the gas phase [33], although to our knowledge this is the first reported detection of short-chain PFCA and PFSA complexes with BSA by nanoESI-MS. Several studies have analyzed intact PFOA– and PFNA–protein complexes via ESI-MS to elucidate the stoichiometry of binding [12,23,26,28,34]. Although caution must be taken when quantifying gas-phase affinities using this method [23] for comparison with solution-based results, the technique has been increasingly used to determine stoichiometries of specific binding for protein–ligand complexes.

Representative deconvoluted mass spectra for 1:1 mole ratio mixtures of BSA and several PFAAs are displayed in Figure 2 and Figure S3. Deconvoluted spectra exhibit distinct peaks for at least one bound PFAA per BSA. Despite somewhat broad peaks, the mass increment of noncovalent PFAA–BSA complex peaks (P + jL) from the initial protein peak (P) generally corresponds well with the theoretical molecular mass of the ligands (ΔM, Table S3). Protein-bound complexes were also maintained for C_2 to C_4 PFCAs in the gas phase, with an observable change in the native BSA spectrum even with the addition of the C_2 PFCA. Peaks corresponding to two bound perfluoropropionate (PFPrA) and perfluorobutanoate (PFBA) (P + 2L) were visible as shoulders at the expected BSA + 2(PFAA) complex mass on the broader peaks. Adduct ions in the protein spectra, which may be partially attributed to contamination of the protein with low-molecular-weight cations (e.g., sodium) via contact with laboratory glassware, may mask the binding of small or weakly bound ligands and also limit evaluation of binding at low ligand:albumin mole ratios. For higher-molecular-weight PFAAs, the pattern of native BSA was preserved with the addition of successive PFAA ligands, leading to a repetition of the native BSA peak shape at intervals corresponding to the mass of each bound PFAA. Evidence of multiple PFAAs bound per BSA molecule was distinctly visible for PFNA and the C_4–C_8 PFSAs (2–3 PFAAs bound at a 1:1 mole ratio). A rough calculation previously indicated that one PFOS binds per albumin molecule [26]. However, we present

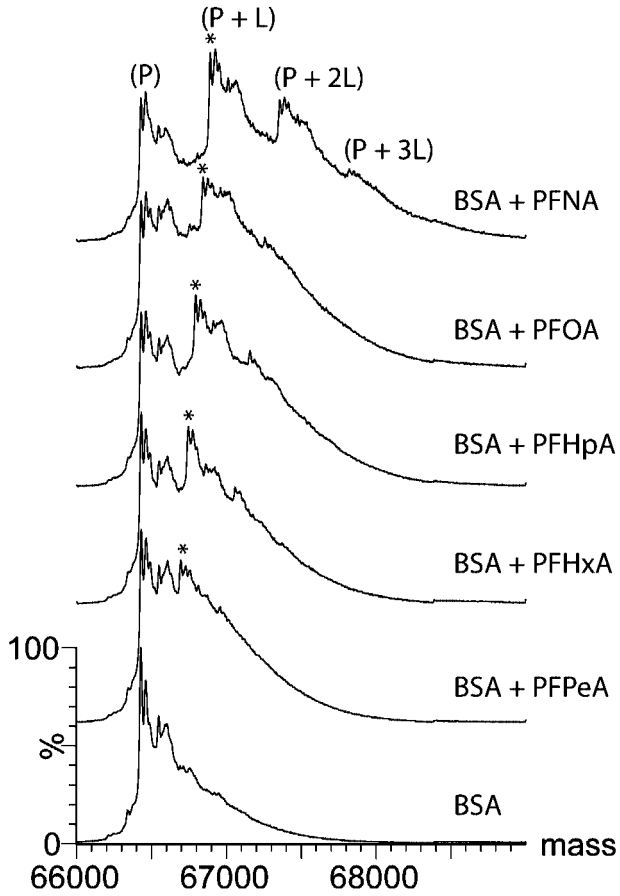


Fig. 2. Deconvoluted spectra of 50 μM BSA alone or with 50 μM PFPeA, PFHxA, PFHpA, PFOA, or PFNA. The location of free BSA (P) and successive bound PFNA ligands (P + L), (P + 2L), and (P + 3L) are denoted on the uppermost spectrum. The peak of each first bound PFAA used to confirm the expected incremental mass shift (ΔM) from the free BSA peak is indicated by a star (*). Analyte acronyms are listed in Table 1.

clear evidence of multiple associations of PFOS with a given albumin molecule at a 1:1 PFOS + BSA mole ratio. Binding sites detected in this manner likely represent both strong and weak associations at the relatively high ligand:albumin mole ratio tested. Lower numbers of C_4 and shorter PFCAs bound at equivalent mole ratios may indicate that secondary binding sites or nonspecific interactions for these compounds, experienced at higher ligand concentrations, are weaker than comparable interactions of longer-chain PFAAs with BSA.

Because the precise binding location for various chain-length PFAAs on albumin has not been established, the full albumin molecule was used in this study to establish specific binding of the full range of PFCAs and PFSAs to BSA, especially for short-chain PFAAs, without selecting for particular domains on the protein. In three homologous domains, albumin contains seven fatty acid (FA) binding sites, including four low-affinity sites (Sites 1, 3, 6, and 7) and three high-affinity sites (Sites 2, 4, and 5) that are likely candidates for PFAA binding [16]. Existing evidence suggests that PFHxA binds to Sudlow's drug binding site II, which overlaps with FA Sites 3 and 4, whereas PFOA preferentially binds to Sudlow's drug binding site I at FA Site 7 [24]. Use of truncated albumin, such as recombinant HSA domain II [35], in future work could improve nanoESI-MS mass resolution and, if coupled with newly developed competitive binding nanoESI-MS techniques [35], may provide further information on site-specific associations.

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Effect of ionic head group on PFAA-BSA binding

The protein–water distribution coefficient determined for PFBS was significantly greater than that for the equivalent fluoroalkyl chain length carboxylate, PFPeA. This corresponded to an association constant for PFBS ($K_a = 7.0 \pm 3.6 \times 10^6$/M) that was more than three times greater than that for PFPeA ($K_a = 2.0 \pm 0.8 \times 10^6$/M). Association constants (ratio of bound to free reactants) were calculated using a one-class binding model that relates the ratio of the bound PFAA concentration to the total protein concentration, $\bar{\nu}$, to the association constant, K_a, the free ligand concentration, and the total number of binding sites on the protein [23]. At the concentrations tested, PFBS and PFPeA exhibited nonlinear isotherms, where a greater fraction of PFBS was bound to BSA relative to PFPeA over the full range tested in dialysis experiments (Fig. 3).

NanoESI-MS was exploited to further evaluate the effect of the ionic head group on PFAA-BSA binding. The stability of two pairs of compounds of equal fluorinated alkyl chain length (4 and 8) and a sulfonate or carboxylate moiety was observed by collision-induced dissociation at 2:1 PFAA:BSA mole ratios. Collision-induced dissociation provides a relatively rapid way to probe the stability of binding in the gas phase. Studies have suggested a link between the energy required for dissociation of gas-phase complexes and solution-phase binding constants, although the relative stability of protein–ligand complexes in the gas phase and in solution influences the interpretation of such measurements [36]. For example, although binding in a hydrophobic pocket may minimize the effects of water on protein–ligand binding in solution [37], hydrophobic interactions may not persist in the gas phase, and desolvation of the protein may have unknown influences on ligand interactions [36].

A markedly different result was observed for PFBS and PFOS compared with PFPeA and PFNA: the PFSAs did not dissociate from BSA over the range of tested collision energies (CE), whereas PFCAs of identical chain lengths dissociated from BSA between 40 and 60 eV (Fig. 3). Representative deconvoluted mass spectra for PFPeA and PFNA at the 2:1 PFAA:BSA mole ratio and spectra demonstrating dissociation of PFNA with increasing CE are displayed in Figure S4. The larger size of the sulfonate moiety relative to the carboxylate may play a role in explaining the different behavior of the PFSAs and PFCAs. The PFAAs may interact with positive residues on BSA, including lysine (pKa 10.53) and arginine (pKa 12.48) [38], and electrostatic interaction between two oppositely charged molecules may be enhanced in the gas phase and become relatively difficult to disrupt when complexes are desolvated [36]. Additionally, BSA and BSA-PFAA complexes were monitored in positive ionization mode, so the dissociation of PFAAs in an anionic form is not expected. Although both PFCAs and PFSAs are expected to be charged at physiological pH in solution, the carboxylate anions may be more amenable to the capture of a proton as compared with the sulfonate before dissociation from BSA in the ESI source and collision cell. Solution-phase methods may be more appropriate to provide insight into the potential contribution of electrostatic interactions to PFAA-BSA binding.

Effect of pH on PFAA-BSA binding

The contribution of electrostatic interactions to protein–ligand binding is often assessed by changing the pH of the solution [39]. However, the analysis and interpretation of the influence of pH on ligand interactions with BSA is complicated by the fact that BSA undergoes conformational changes with changes in pH, potentially altering binding site characteristics. Conformational changes can occur through the folding and unfolding of the tertiary and secondary structure of the protein. In the case of BSA, successive multi-step, reversible transformations occur with increasing pH, from fast (F) to normal (N) near pH 4, and to basic (B) and aged (A) forms near pH 8 and 10, respectively. At near neutral pH (pH 7.4), albumin has a net charge of −17 and is heart-shaped in the N form [15,40]. Over the pH range tested (pH 6–9), BSA begins in the N form at lower pH and transitions to the B form. An analysis of the native fluorescence of BSA reveals a decrease in fluorescence intensity with increasing pH and a slight blue shift at pH 9 (Fig. S5). These spectroscopic changes confirm the conformational changes in BSA as a function of pH.

For all tested PFAAs, equilibrium binding to BSA was high across all tested pH levels, ranging from 85% bound for PFPeA at pH 9 to greater than 99% for several compounds and pH conditions (Table S4). Similar to K_{PW} results described earlier, at pH 7 the average number of PFCA molecules bound per BSA molecule ($\bar{\nu}$) increases for the C_4 to C_6 compounds and

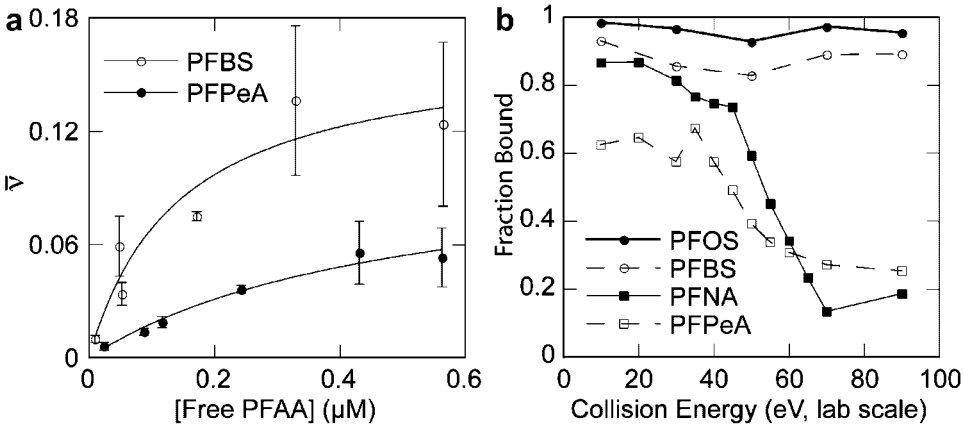


Fig. 3. Effect of ionic head group on binding of equivalent-chain-length PFAAs to BSA. In equilibrium dialysis tests (**A**), PFBS (white) exhibits higher affinity for BSA than PFPeA (black) across the full concentration range tested. Data are displayed as the ratio of the bound PFAA concentration to the total protein concentration ($\bar{\nu}$) versus the free PFAA concentration and were fit with a one-class binding model to determine K_a ($R^2 = 0.906$ and 0.973 for PFBS and PFPeA, respectively). The measured fraction of PFAAs bound to BSA in nanoESI spectra (**B**) decreased with increasing collision energy for PFCAs (black and white squares) at 40 to 60 eV, whereas PFSA-BSA (black and white circles) complexes did not dissociate over the range of collision energies. Analyte acronyms are listed in Table 1.

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decreases above C_8 (Fig. S6). No increase was observed in bound PFSAs with increasing chain length at pH 7. A plot of the dialysis bag concentration versus reservoir pH reveals an overall increase in binding of PFBS, PFPeA, and PFHxA to BSA with pH, whereas a negative relationship was obtained for several longer chained PFAAs: PFHxS, PFOS, PFUnA, and PFDoA (Fig. 4 and Table S5). Little to no overall trend in binding with pH was observed for PFHpA, PFOA, PFNA, and PFDA (Fig. S7). The opposite trend in $\bar{\nu}$ observed for short- and long-chain PFAAs with pH may result from different conformational changes experienced from binding at different locations on the protein.

Fluorescence analysis was used to further probe the binding of PFNA and PFOS—two PFAAs with equivalent chain length but different ionic head groups—to BSA. The binding of PFAAs to serum albumin evokes changes in the protein's native fluorescence (Fig. S8); these spectroscopic changes may be used to estimate binding constants and stoichiometries, as recently described [17]. For both PFAAs, the estimated binding constant, K_{Hill}, determined from a plot of the degree of saturation (Y) versus total PFAA concentration (Fig. S9), increases from pH 6 to pH 7, but no further change is observed at higher pH (Fig. S10). This trend is similar to that obtained for these PFAAs by equilibrium dialysis (Figs. 4 and S7); however, because the fluorescence data are obtained at higher PFAA:BSA mole ratios, these data are not directly comparable. Given the changes in native fluorescence of BSA as a function of pH, the differences in binding affinity observed here are likely attributable to conformational changes in the protein. However, electrostatic changes could also play a role.

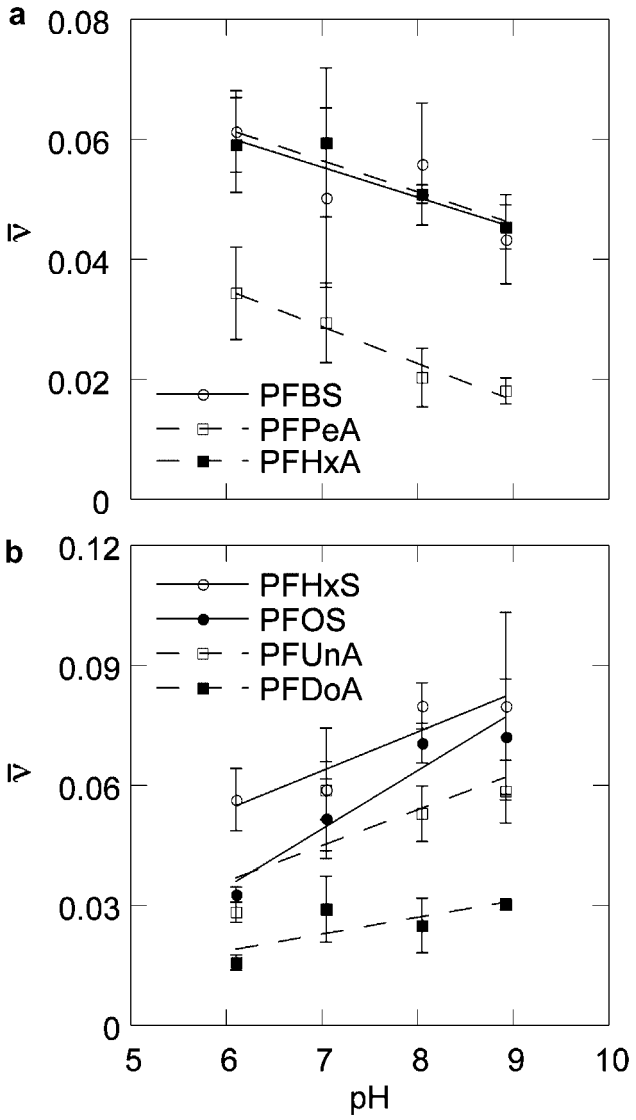


Fig. 4. Effect of pH on $\bar{\nu}$, the concentration of PFAA bound to BSA normalized to the total protein concentration. Binding of PFBS (white circles), PFPeA (white squares), and PFHxA (black squares) to BSA (**A**) decreased with increasing pH, whereas binding of PFHxS (white circles), PFOS (black circles), PFUnA (white squares), and PFDoA (black squares) (**B**) increased with pH. Analyte acronyms are listed in Table 1.

One would expect enhanced electrostatic interactions with decreasing pH as repulsion from negatively charged amino acid residues is reduced; however, BSA supports a more compact conformation as it approaches its isoelectric point (~pH 4.8), which may reduce accessibility of surfactants to the hydrophobic cavities of the binding sites [41]. Gelamo et al. [41] observed a lower binding constant for sodium dodecyl sulfate with BSA at pH 5 as compared with pH 7 and pH 9. The compact structure may have a more pronounced effect on binding of the longer, more hydrophobic PFAAs, whereas electrostatic interactions may play a larger role in binding for short-chain PFAAs. The difference in responses of PFAA—BSA affinity with pH suggests that short- and long-chain PFAAs may bind at different locations on albumin.

To specifically investigate the role of electrostatics, the binding of PFNA and PFOS to BSA as a function of ionic strength was explored through fluorescence titrations by adding varying concentrations of sodium chloride to the titration solutions. The results are shown in Figures S11 and S12. For both PFAAs, no difference was seen in estimated binding affinity over a physiologically relevant ionic strength range (Fig. S13). This result supports the hypothesis that the observed changes in PFAA-BSA binding with changing pH result primarily from conformational changes in the protein and not changes in electrostatics.

Significance

In recent attempts to include protein associations in bioaccumulation models, which traditionally approximate organism or tissue sorptive capacity using only lipid content, researchers used BSA–ligand relationships to report log K_{PW} values for a range of hydrophobic organic contaminants [14]. Empirical data for log K_{PW} of a series of PFAA cogeners are reported in the present study, yielding relatively large values from 3.3 to 4.3. Considered analogous to octanol–water partition coefficients, such parameters may be useful to more accurately predict chemical accumulation and distribution. However, use of K_{PW} to describe binding between macromolecules and small molecule ligands should be conducted with caution and limited to a narrow concentration range over which binding may be approximated as linear.

Results from the present study indicate increased binding with chain length for C_4 to C_6 PFCAs. Above C_8, K_{PW} decreased with increasing chain length. Additionally, differences in binding of PFBS and PFOS relative to PFPeA and PFNA, respectively, observed via dialysis and collision-induced dissociation indicate an electrostatic component to interactions with BSA. Fluorescence results for PFOS and PFNA suggest that these effects are minimal in solution. A number of studies have illustrated an important role of the hydrophobic driving force on PFAA environmental partitioning. However, increased rigidity associated with long-chain PFCAs may contribute to the

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observed nonlinear relationship of K_{PW} with the fluorocarbon tail length.

Prior studies suggest that the binding of short-chain PFAAs may be different in nature than that of long-chain PFAAs [24,26]. Perfluorooctanoate and PFHxA may bind primarily to different sites on HSA, demonstrating that an increase in the fluorinated carbon tail length by only two units may have a substantial effect on the nature of PFCA–albumin interactions [24]. Fluorescence studies indicate that long-chain PFAAs have a greater influence on albumin conformation than short-chain PFAAs (e.g., PFBS, PFPeA, and PFHxA) [17,18], and the Trp binding site on HSA, which overlaps with or exists in the hydrophobic cavity of fatty acid site 7 in subdomain IIA, may have a preference for long-chain PFAAs [16]. We find strong binding of short-chain PFAAs to BSA, suggesting that reduced hydrophobicity and steric hindrances of short-chain PFCAs and PFBS, which may limit observable conformational changes in albumin by fluorescence methods, do not correspond to low affinity for albumin. In the present study, pH-induced changes in binding affinity observed via equilibrium dialysis support evidence that short- and long-chain PFAAs bind at different locations on BSA. Fluorescence titrations suggest that the observed pH dependence of binding is attributable to conformational changes in the protein.

An effort to reduce the bioaccumulation of PFAAs in humans and wildlife has led to shifts in fluorochemical production to shorter-chain-length compounds. Association constants obtained for PFBS and PFPeA are useful to compare among various PFAAs and with other exogenous and endogenous ligands in blood. Results in the present study indicate that short-chain PFAAs bind strongly to BSA at low PFAA:albumin mole ratios, suggesting that physiological implications of strong binding to albumin may be important for short-chain PFAAs. However, these results contrast with limited evidence that short-chain PFAAs are less bioaccumulative than long-chain PFAAs [5,6], highlighting a need for additional protein association measures for bioaccumulation modeling beyond a single-protein K_{PW}. Further research is needed to investigate the binding of short-chain perfluorinated compounds to a multitude of potential protein and transporter targets and to further understand the influence of such interactions on biological uptake, retention, and functioning.

SUPPLEMENTAL DATA

Figures S1–S13.
Tables S1–S5. (1,103 KB DOC)

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